

Ming (Simon) Chen

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EDUCATION

University of Alberta

Bachelor of Science in Computer Science, Specialization

June 2024

Edmonton, AB

AI & TECHNICAL SKILLS

AI Tools: ChatGPT Codex, Claude Code, Google Gemini, MCP, RAG, embeddings, vector databases, semantic search

Languages: TypeScript, JavaScript, Python, C++, Java, C#, SQL, HTML, CSS

Frontend: React, Angular, Next.js, React Native, Tailwind CSS, responsive UI, accessibility

Backend & APIs: Node.js, FastAPI, REST APIs, GraphQL, microservices, asynchronous programming, API design

Cloud & Infrastructure: AWS Lambda, API Gateway, S3, DynamoDB, CloudWatch, Docker, Kubernetes

Data & Distributed Systems: PostgreSQL, MySQL, MongoDB, Redis, pgvector, data modeling, caching, Kafka

EXPERIENCE

AI Software Developer

Sep 2024 – June 2026

YOYA.COM

Xiamen, Fujian, China

- Developed AI-assisted features for an education content platform, helping teachers and students create animated lessons, interactive courseware, and multimedia learning materials while reducing content-production time by 40%
- Integrated large language model capabilities for prompt-based content generation, including scripts, lesson ideas, educational copy, and structured creative workflows, cutting initial drafting time by 50%
- Built retrieval-augmented workflows using embeddings and semantic search to connect AI-generated content with reusable education resources and platform knowledge, improving content retrieval relevance by 25%
- Built an animated avatar companion with text-to-speech voice capabilities that responds to user inputs through conversational interaction and animation, supporting 20+ interactive dialogue flows
- Developed a low-latency, on-device generative system using Ollama and Qwen for text generation, voice TTS, and real-time generative animation with sub-second response targets

Full-Stack Software Engineer

Jan 2022 – Sep 2022

Titan Logix Corp.

Edmonton, AB

- Built and maintained a production Angular and TypeScript dashboard for 1,500+ active users, improving page load performance by 35% through component optimization and API caching
- Developed AWS serverless services with Lambda, API Gateway, DynamoDB, S3, and Serverless Framework CI/CD workflows, supporting 2M+ monthly API requests with 99.9% uptime
- Designed REST APIs and telemetry data pipelines connecting 10,000+ IoT devices for real-time truck sensor monitoring, diagnostics, anomaly review, and fleet data visualization
- Processed high-frequency sensor data for pressure, level, and temperature readings, improving liquid level accuracy by 28% through data quality checks

Artificial Intelligence Researcher

May 2021 – Sep 2021

University of Alberta

Edmonton, AB

- Researched code synthesis for generating stronger heuristic functions before LLM coding assistants became mainstream, improving combinatorial search efficiency by 22–35%
- Built Python evaluation pipelines for Dijkstra, A*, IDA*, and Metropolis-Hastings across 1,000+ sliding-tile puzzle instances to validate synthesized heuristic quality
- Scaled multiprocessing experiments on Compute Canada HPC clusters, applying early program-synthesis ideas now central to agentic AI coding workflows

Frontend Software Engineer

May 2020 – Sep 2020

YOYA.COM

Xiamen, Fujian, China

- Developed a WeChat Mini Program serving 50K+ monthly active users, shipping 20+ mobile features with WXML, WXSS, JavaScript, and API integrations
- Built 30+ reusable UI components through code review, testing, and debugging, cutting delivery time by 40% while improving interface consistency
- Integrated WeChat APIs for authentication, payment, QR scanning, and location services, contributing to a 25% checkout conversion lift

PROJECTS

2D Game Engine

C++, SFML, Dear ImGui, Entity Component System

github.com/schen32/IsometricTerrainGeneration

- Built a custom 2D game engine in modern C++ with SFML rendering/input, Dear ImGui debugging tools, and modular gameplay systems
- Architected a data-oriented ECS supporting 10,000+ active entities per scene with reusable rendering, movement, collision, and state-update systems

Arduino Restaurant Maps

C, Embedded Systems, Arduino

github.com/schen32/Restaurant-Finder

- Built a C-based Arduino touchscreen interface with 4-direction joystick navigation, implementing screen-state management, menu controls, and interactive map rendering across 6 city regions
- Implemented an embedded restaurant directory containing 20+ locations with name, address, category, and rating data, enabling users to select map markers and view detailed venue information